The Incidence of Flood Risk

Thomas Bézy*

November 2023

Preliminary and incomplete

Abstract

This paper documents the vulnerability and exposure to flood risk with fine-grain data for the universe of homeowners, landlords and tenants in France. Using data at the dwelling level, I distinguish the exposure of owner-occupied, rented, second, and vacant homes, and link these properties to the disposable income and housing wealth of their owners and tenants. Vulnerability to floods exhibits large heterogeneity. Tenants and homeowners owning only their primary residence are highly vulnerable due to limited housing wealth, whereas most landlords diversify their assets, exposing only a fraction of their housing wealth to flood risks. In addition, I find that high-income homeowners and tenants are underrepresented in flood-prone areas. As a result, the income of homeowners is 6% lower in risky areas, and the income of tenants is 9% lower. On the contrary, second homes owned by middle and high-income households are overrepresented in flood-exposed areas, with a twofold higher likelihood. These patterns of residential distribution appear to be driven by high-income homeowners and tenants actively avoiding flood risks and by an amenity effect influencing second home owners. I show that these findings matter for adaptation policies: (1) The cost of buying-back homes at risk could be divided by five if it focused on low-income homeowners. (2) A subsidized insurance policy such as the current French CatNat system redistributes 5% of total annual losses caused by floods from the bottom 50% to the top 50% through regressive premiums, because of poor targeting toward the most vulnerable households.

Keywords: Wealth Inequality, Insurance, Natural Disasters, Sorting

JEL Codes: D31, G52, Q51, Q54

*Paris School of Economics (email: thomas.bezy@psemail.eu), website: https://thomas-bezy.github.io/Website/. I am deeply grateful to Katrin Millock and Lucas Chancel for their invaluable guidance, insights, and encouragement throughout this project. I would like to thank Philipp Bothe, Carolyn Kousky, Antoine Levy, Romaine Loubes, Thomas Piketty, Stefanie Stantcheva, Swenja Surminski, as well as seminar participants for very helpful discussions and suggestions. Data used in this project was in restricted access and obtained through the *Centre d'Accès Sécurisé aux Données* (CASD). This work was funded by a French government subsidy managed by the Agence Nationale de la Recherche (ANR) under the framework of the "*Investissements d'avenir programme*" reference ANR-17-EURE-001.

In 2021, the economic cost of natural disasters reached 343 billion dollars worldwide, and two developed countries, Germany and Belgium, set all-time records for their most costly weather-related disasters (Masters, 2022). The economic burden of these natural disasters is anticipated to surge (Hallegatte et al., 2013; Smith and Katz, 2013). The escalating frequency and severity of natural disasters play a crucial role in accounting for this upsurge. In addition, the ongoing economic expansion of urban centers (Balboni, 2019; Kocornik-Mina et al., 2020) and the continuous growth of seaside resorts (Châtel et al., 2021; Smith, 2021) may lead to a substantial surge in both the number of households exposed to risk and the overall value at risk. This phenomenon poses a crucial question: who bears the economic consequences of natural disasters?

Previous evaluations of natural disaster incidence utilized aggregate income data, considering the income of residents (Walker et al., 2006; Sayers et al., 2018; Bakkensen and Ma, 2020; Rözer and Surminski, 2021; Wing et al., 2022). However, this approach presents two significant limitations. First, it conflates the exposure of homeowners and tenants, who may exhibit distinct vulnerability levels and sorting patterns. Second, from the results of this paper, it appears that 60% of dwellings in exposed areas are owned by households residing outside these vulnerable locations. Consequently, assessing owners' exposure cannot rely on aggregate income and wealth data.

In this paper, I make use of novel administrative data (André and Meslin, 2021) that provides the exact location of every dwelling in France and links them with information on the income and wealth of owners and tenants. This unique dataset covers all types of dwellings, including owner-occupied, rental, second, and empty homes. The last two categories are often overlooked in the literature, and rental homes are generally not matched to their owners. I overlap these data with exposure maps for flooding, which is the costliest type of event in France and in Europe. I then compute the number of dwellings at risk and their value, categorized by owners' income and wealth levels. This approach enables me to measure how owners diversify their real estate wealth when facing the threat of being flooded and to evaluate sorting patterns over flood risk. Finally, I derive implications of these findings on the incidence of two adaptation policies: home buyouts and insurance against extreme weather events. This study provides the first analysis of owners' and tenants' exposure to floods, differentiating them by income and wealth at the most granular level possible.

I focus on expected exposure derived from exposure maps rather than past damages for two main reasons. The first reason is that expected risks appear to be more relevant for policy

design, as exposure maps serve as the best available predictor of future exposure. Past damages resulting from natural disasters can be influenced by rare, unpredictable, and large-scale events, such as the Alex storm in the *Vallée de la Roya* in France in 2020. Therefore, considering past event damages to assess exposure would give undue weight to these large-scale events, which are unlikely to be representative of future disaster damages. The second reason is that exposure maps provide a finer-grained level of detail, allowing me to capture variations at the dwelling level.

First, on the owners' side, I find that floods have large heterogeneous impacts due to significant differences in asset diversification across wealth and income distributions. For each owner, I compute the share of housing value at risk of being exposed to flood risk. I find that 40% of exposed owners have their entire wealth at risk, which implies that their whole housing wealth is in jeopardy due to flooding. In contrast, 40% of exposed owners have less than half of their total housing wealth at risk. Additionally, 10% of exposed owners have less than 10% of their housing wealth exposed to flooding. These findings indicate that the widely held belief that natural disaster victims are "left with nothing" holds true to some extent. For 40% of exposed owners, a highly violent flood would indeed result in a complete loss of their housing wealth. In the large majority of cases, these households are homeowners owning only their place of residence and no other housing wealth. However, for the majority of owners, a similar event would only lead to the partial destruction of their wealth, with less than half of their wealth being at risk in 40% of cases. These findings underscore the importance of distinguishing between the types of dwellings affected, as the impact of having an owner-occupied home flooded differs significantly from having a rented, second, or empty home underwater. Finally, tenants remain the most vulnerable households as most of them only own their furniture which are very much exposed to flooding. Only 15% of them own housing wealth and 28% earn financial income, highlighting an extremely low degree of diversification.

Second, owner-occupied homes are less prevalent in exposed areas, comprising only 40% of the dwellings at risk, despite representing 55% of all dwellings in France. Additionally, high-income homeowners and tenants sort less over flood risk areas, resulting in average incomes for homeowners and tenants that are respectively 6% and 9% lower in exposed areas. This observation suggests that high-income households tend to avoid residing in high-risk areas when selecting their place of residence.

Third, second homes owned by middle- to high-income households, which constitute less than 10% of all residential properties in France, represent 20% of the dwellings at risk. Interest-

ingly, when analyzing the sorting patterns of second homes by income group, it appears that the degree of overexposure to flood risk decreases after the eighth decile. This suggests that individuals at the upper end of the income distribution are less inclined to invest in second homes located in high-risk areas, showing a preference for alternative locations, such as urban or mountainous areas.

Fourth, the combination of these two effects, namely the higher exposure of low-income homeowners and middle- to high-income owners of second homes, tends to balance each other out, resulting in an overall income distribution of owners in exposed areas that closely resembles the national income distribution. Nevertheless, as low-income households bear a greater exposure through their primary residences, whereas high-income households are exposed through their second homes, the repercussions of flooding are considerably more profound for low-income households. These findings illustrate that high-income households possess a greater capacity to diversify their assets in the face of flooding risks, whereas low-income households more frequently find themselves with both their primary residence and their entire wealth concentrated in areas vulnerable to flooding.

Fifth, these patterns still hold when examining within-municipality variations, suggesting that they cannot be fully explained by differences in amenities between municipalities. The under-exposure of high-income homeowners and tenants could be attributed to a strategic avoidance of flood risk. In contrast, the over-exposure of second homes can be explained by within-municipality amenities, such as the direct view of the coast. I plan to investigate these mechanisms in-depth in further analysis.

Finally, I draw implications from these findings for two adaptation policies. The first policy relates to home buyouts, in which the government acquires flooded properties for demolition. Typically, these initiatives are event-driven and do not specifically target income groups based on their vulnerability to flood impacts. I simulate an ambitious policy that involves repurchasing all at-risk homes. My analysis reveals that if such an initiative were to prioritize homeowners in the bottom 50% of the income distribution, it could potentially reduce buyout costs by five, highlighting substantial cost-saving opportunities when governments take economic vulnerabilities into account while designing buyout programs.

I also analyze the implications for the subsidized French insurance against extreme weather events. This type of insurance scheme is gaining in popularity in recent years, in response to the significant undercoverage observed in certain countries such as the U.S. (Wagner, 2022). In

France, this insurance system is funded through a mandatory, uniform rate additional premium[1] imposed on all households across the nation to provide coverage for natural disaster victims. First, I demonstrate that the fee structure exhibits regressive characteristics. Second, given the previous results, this insurance system effectively compels all households in France to contribute towards insuring a substantial portion of others who may not necessarily require insurance in priority, particularly second-home owners. I propose potential strategies to enhance the financing of such insurance schemes, which could be valuable for policymakers willing to implement similar systems in other countries.

Related literature. My work intersects with several existing bodies of literature. Firstly, it aligns with the expanding field of environmental justice (Taylor, 2000; Mohai et al., 2009; Banzhaf et al., 2019), which seeks to comprehend the unequal distribution of environmental risks and the underlying reasons. Specifically, my research extends the prior work on sorting patterns over flood risk. In the U.K., utilizing aggregate income data, studies by Walker et al. (2006), Sayers et al. (2018), and Rözer and Surminski (2021) have observed that low-income households tend to be more frequently situated in flood-prone areas. Similar results were found for Germany by Osberghaus (2021) using survey data. Furthermore, there is a burgeoning literature on this topic in the United States. Wing et al. (2022) found that the costs of natural disasters disproportionately impact poorer communities, which also tend to have a higher proportion of White residents. Bakkensen and Ma (2020) discovered that low-income and Hispanic households tend to cluster in high-risk areas. Both studies used aggregate census data to assess exposure by income group. However, Sastry (2022), using transaction-level data, found evidence of credit-rationing in flood-prone areas, discouraging credit-constrained borrowers from residing along the coast and resulting in higher exposure for high-income property owners. Consequently, the majority of the literature, which relies on aggregate income data, suggests greater exposure for low-income households. In contrast, Sastry (2022), focusing on homeowners and utilizing transaction-level data in the specific context of Florida, identified evidence of sorting behavior over flood risk among high-income households. My contribution to this literature is data-driven and distinctive in two key ways: it distinguishes between the exposure of property owners and tenants, a differentiation not feasible with aggregate income data, and it differentiates by type of dwelling, encompassing owner-occupied, rental, second, and empty

[1] This additional premium represents 12% of housing insurance contracts and 6% of car insurance contracts.

homes. The latter two categories are often overlooked in most of the papers cited above, despite representing 30% of the dwellings at risk in my analysis and implying markedly distinct vulnerability levels.

My work also builds upon prior research concerning wealth inequality dynamics within the housing market. In France, the wealth held by the bottom 50% of the population accounts for just 5% of the total national wealth (Chancel et al., 2022) and is mainly composed of real estate assets (Fagereng et al., 2020; Bach et al., 2020). For low-income homeowners who lack any other form of wealth aside from their primary residence, diversifying their assets is often exceedingly challenging and, in most cases, unfeasible. This lack of diversification among those at the lower end of the wealth spectrum exposes small homeowners to a heightened level of risk from large-scale climate-related events. My contribution to this body of literature is empirical in nature, involving the measurement of the varying degrees of wealth vulnerability to floods across the spectrum of property owners.

This paper is closely connected to the literature on insurance against extreme weather events. One prominent thread in this literature centers on insurance affordability and its associated advantages. Studies such as Kousky and Kunreuther (2014), Grislain-Letrémy (2018), and Hudson (2018) have examined this issue within the contexts of the U.S., French overseas departments, and Europe, respectively. My work directly contributes to this literature, particularly in the case of the French CatNat system, which was initially designed to render insurance more affordable. Nevertheless, I argue that affordability alone may not sufficient for an insurance system to be equitable, drawing attention to the example of the CatNat fee, which results in insurance premiums lower than those in other countries but exhibits regressive characteristics.

Finally, another significant segment of the literature examines the redistributive implications of insurance against natural disasters. Studies like Holladay and Schwartz (2010) and Bin et al. (2017) assess the distributional effects of the U.S. National Flood Insurance Program (NFIP) and find that the current U.S. system is slightly regressive. Additionally, Ben-Shahar and Logue (2016) and Dinan et al. (2019) observe that the NFIP transfers income from a multitude of inland areas to a minority of coastal ones. More closely aligned with my research context, Charpentier et al. (2022) highlight that flood risk is considerably concentrated in France, with 10% of households bearing 74% of the losses. They also underscore the high exposure to extreme weather events in affluent areas, implying that the French insurance system may primarily benefit wealthier households. My contribution to this literature involves the empirical measurement, leveraging unique fine-grained data, of how each category of property owners actually

benefits from a government-subsidized insurance system such as CatNat.

The remainder of the paper is structured as follows. In Section 1, I provide some context on flood risk in France and how it is covered by insurance. I describe the data in Section 2. I provide descriptive results on the heterogeneity in vulnerability to floods in Section 3. In section 4, I study the patterns of sorting over flood risk and their mechanisms. Finally, in section 5, I draw conclusions for the incidence of adaptation policies and conclude in Section 6.

1 Flood Risk in France

Between 1980 and 2022, floods accounted for nearly 43% of climate-related economic damages in Europe, amounting to 280 billion euros over the period and establishing themselves as the most financially burdensome type of event on the continent (Commission, 2022). In France, according to the 2021 report from the French insurance association (France Assureurs, 2021b), floods were also responsible for the highest amount of damages, with an annual average of 1 billion euros from 1989 to 2020. Projections indicate that losses are anticipated to increase by 87% over the 2020-2050 period, reaching approximately 1.8 billion euros annually. Of this increase, 65% can be attributed to a higher number and value of properties at risk, while climate change factors would account for only 26% of the rise. This underscores the importance of gaining a better understanding of how individuals position themselves in relation to flood risk to mitigate the impacts of climate change. It's noteworthy that a majority of the losses, specifically 68% of total losses, are concentrated in residential dwellings, in contrast to commercial buildings which account for the remaining 32% of total losses (CCR, 2022). For the remainder of this study, our focus will be exclusively on residential real estate.

On average, among all insured dwellings, approximately 0.5% to 1% are victims of flooding each year (France Assureurs, 2021b). The average cost of a residential dwelling insurance policy stands at 10,000 euros, exclusive of the deductible[2]. In Germany, the average building flood insurance claim is twice as high as the average content-related claim (Osberghaus,

[2]The deductible is determined by law. If a municipality experiences its first flood, the deductible is 380 euros. Subsequently, for municipalities that undergo multiple flood events without implementing a risk protection plan (PPR) aimed at documenting the location of risks and urban development, the deductible increases. Specifically, after the third event without a PPR, it doubles, after the fourth it triples, after the fifth it quadruples, reaching 1520 euros. For more details, refer to the *Mission risques naturels* website. Among the 34,965 municipalities in France in 2021, 505 had their deductibles doubled, 127 had them tripled, and 22 had them quadrupled (France Assureurs, 2021b).

2021). Unfortunately this specific figure is not available for France. Applying these proportions to France means the average building insurance claim (benefiting the owner) would be 6,667 euros, equivalent to around 5% of the value of dwellings situated in municipalities exposed to floods between 2017 and 2019. On the other hand, the average content-related claim (benefiting the tenant, if any) would be 3,333 euros.

France is also characterized by a unique insurance system against natural disasters, established in 1982 and referred to as "CatNat" [3]. This system mandates that private insurance companies incorporate coverage for natural disasters into their standard automotive and property insurance policies. The funding mechanism involves a mandatory additional fee, constituting 6% of the initial premium for car insurance and 12% for housing insurance. Of the contributions collected, 88% is channeled to the "Caisse Centrale de Réassurance" (CCR), a reinsurance company wholly owned by the French government, with the remaining 12% directed to the "Fonds de Prévention des Risques Naturels Majeurs," also known as the "Fonds Barnier." This fund is specifically allocated for the acquisition of properties situated in high-risk areas.

When a disaster occurs, the municipality is required to submit a CatNat form for review. This evaluation is conducted by a committee, which includes representatives from the CCR, as well as members from the Ministries of the Economy and the Environment. If the application is not rejected, the municipality receives the CatNat designation, and insurance companies then provide compensation to the victims of the disaster. Ultimately, the CCR acts as a reinsurer for these insurance companies, covering the costs they have incurred[4].

This arrangement ensures that French households can access insurance at an affordable price, resulting in a 98% uptake of housing insurance throughout the country. Insurance companies are also largely aligned with the CatNat system, with the CCR commanding more than 90% of the market share in the natural disaster reinsurance sector (CCR, 2022). Without such a program, the insurability of extreme weather events would be severely compromised due to the exorbitant cost and the intricate spatial correlation of such events, rendering diversification considerably more complex for private insurance companies (Jaffee and Russell, 1997). Furthermore, since insurance companies bear only a small portion of the overall cost, they tend not to significantly adjust premiums based on the risk of exposure (Sansévérino-Godfrin, 1996; Grislain-Letrémy and Peinturier, 2010; Grislain-Letrémy, 2018).

[3]See McAneney et al. (2016) for a review of alternative government-sponsored insurance systems

[4]From 1982 to 2022, the CCR was responsible for 51% of the total claims. During major disasters, such as in 2017, the CCR's share of the disaster-related costs increased to 70% (CCR, 2022).

Apart from the exceptional case in 1999, France has not required government intervention for rebuilding dwellings after extreme weather events since the inception of the CatNat system in 1982. However, the substantial escalation in the cost of natural disasters in recent years has resulted in a significant reduction in the CCR's reserves, with a decrease of 50% from 2015 to 2021 (Cour des comptes, 2022). This situation raises important questions regarding the long-term sustainability of the CatNat system. Consequently, there is a mounting discourse surrounding the potential necessity of augmenting contributions on housing insurance from the existing 12% to 19% (CCR, 2023). We will discuss the incidence of such a reform in Section 5

2 Data

2.1 Households and Dwellings Characteristics

To study the economic characteristics of households exposed to flood risk, I use the 2017 Demographic Database on Housing and Individuals from INSEE[5]. I am able to recover the characteristics of both owners and tenants based on the merging realized by André and Meslin (2021). For owners, I observe the characteristics of mom and pop owners and as well as those who hold real estate through property management companies. However, I do not have the characteristics of owners when real estate is owned by LLCs or institutional investors. Households' characteristics correspond to household size, age, income and structure of income (labour income, net rental income, other financial income, pensions). To build income deciles and percentiles, I consider disposable income by consumption unit following the methodology of INSEE (2023)[6]. The analysis presented in this paper is a cross-section, but I plan to add a panel dimension and enlarge the study to the 2015-2022 period.

This dataset encompasses information on all households within the country and provides details on the characteristics of buildings. These characteristics include the construction date, dwelling type (whether it is a house, apartment, or designated as social housing), floor number, and the type of occupancy (owner-occupied, rental, second home, or empty dwelling). It is important to note that, as per the definition provided by the *French National Institute for*

[5] *Fichiers Démographiques sur les Logements et les Individus 2017*, INSEE. Data are collected through housing and property taxes and are confidential. They must be accessed with an authorization from INSEE and through the *Centre d'Accès Sécurisé aux Données* (CASD).

[6] The value of a consumption unit is defined as 1 for the first member of a household, 0.5 for every additional member in the household and 0.3 for children below 14 years old.

Statistics and Economic Studies (INSEE), the distinction between owner-occupied and second homes is the requirement that a household must spend a minimum of six months per year in an owner-occupied dwelling, whereas second homes are typically used for shorter duration. Second homes are documented in this administrative database, as they can have implications for varying tax schemes, particularly regarding housing taxes in select municipalities. Lastly, an empty dwelling can be attributed to one of three primary reasons: it may be listed for sale or rent on the real estate market, the owner may have yet to pay the estate tax that allows for occupancy, or it may be retained by the owner but remains unoccupied due to its inadequate condition.

In the remaining of the paper, we will refer to three categories of households exposed to floods:

- Exposed homeowners: households having their main place of living at risk.
- Exposed landlords: households owning dwellings at risk in which they do not reside.
- Exposed tenants: households occupying a dwelling rented from a landlord. Tenants can also own a second home or another type of dwelling, but they do not own their place of living.

After excluding non-resident owners, and other private investors not covered in the dataset such as LLCs, I am left with 29 million dwellings out of the 35.7 million dwellings in France in 2017. The dataset thus covers around 80% of the total housing market in France.

Previous studies that relied on aggregate income data encountered limitations as they couldn't distinguish between property owners and tenants; these data sources were grounded in the collective income of residing households but lacked the granularity required to make this distinction. Secondly, this dataset provides an exceptional opportunity to scrutinize flood exposure at an exceedingly granular level – that is, at the level of individual dwellings. The precise geolocation data of each household enables the highly accurate measurement of flood exposure. Finally, this dataset encompasses all categories of dwellings, including second homes and empty properties, which have, in many instances, been neglected within the existing literature, despite constituting a substantial 30% of dwellings in flood-prone regions. Although these dwellings often remain vacant, there is a tangible cost associated with their reconstruction when they are impacted by floods. This is particularly relevant in the context of government-guaranteed insurance, as is the case in France. In such scenarios, the financial burden of repairing these

structures is collectively shouldered by households across the entire country, underscoring the critical need to gain a deeper understanding of the extent to which these dwellings are exposed to risks and the associated patterns of sorting.

Finally, I gather data on house prices at the municipality level from *Données des Valeurs Foncières*. This dataset gives the average annual number of sales and price per meter squared by municipality. Housing value is defined in this paper as the product of the surface of the dwelling and the price per meter squared at the municipality level.

2.2 Exposure to Floods

I study two types of flooding events: river and coastal flooding. Over the 2020-2050 period, they are expected to account respectively for 93% and 7% of total damages caused by floods (France Assureurs, 2021b). The maps of exposure I use are the Flooding Risks Areas (FRAs)[7]. FRAs are the most reliable maps available for France. They account for local particularities such as flood protections and categorize 3 types of risks: frequent (small scale but frequent events, return period of 10 to 30 years), medium (return period of 100 to 300 years) and exceptional (large scale events but extremely rare, return period of 1000 years and above). We will mostly focus on frequent events as they represent the largest share of total losses[8]. FRAs are also available for overseas departments. However, they do not cover the whole country and are available only in specific areas where the risk of flooding is particularly high. After overlaping the maps with the geolocation of dwellings in France, I find that FRA maps cover around 45% of dwellings in France in most exposed areas, which still make these maps still accurate and representative of the overall flooding risk in France. I consider that the remaining 55% of dwellings are not exposed to floods. One can see the area covered by FRAs for river flooding in Figure 2a and for coastal flooding in Figure 2b.

I also exploit the Approximate Envelopes of Potential Flooding maps (AEPF)[9]. These maps have the advantage of covering the whole Metropolitan France. However, they do not cover overseas departments and do not take into account local flood protections. They use a "maximalist" approach, which means that they cover areas that would be affected if a very extreme

[7] *Territoires à Risque important d'Inondations* (TRI) from *Géorisques*.

[8] This information comes from discussions with insurers and *Mission Risques Naturels*. No official number is available in France.

[9] *Enveloppe Approchée des Inondations Potentielles* (EAIP) from the French Ministry of the Environment. AEPF maps are the official maps used on the *French government's website*. The shapefiles used in this paper are not publicly available and were requested to the Ministry of the Environment.

scenario was to happen (superior to a 1000 years return period). Also, they do not include any measure of risk intensity. I use these maps to check for the robustness of the results obtained with FRAs. Results are very robust, but not displayed in this version of the paper. They are available upon request.

In Figure 1, one can observe the difference between the two types of maps. It appears that AEPF maps correspond to exposure to large-scale events, whereas FRAs measure exposure to more frequent and small scale events. Most important, one can notice how FRA High Risk zoning follows flood protection lines, in particular on the upper part of the figure. This illustrates how flood protections are taken into account when modelling FRAs. We can also observe that FRA Middle Risk zoning do not follow flood protection lines anymore, which makes sense as floods have a higher probability of going beyond protections during large-scale events.

2.3 Insurance Spending

To study the redistributive effects of the CatNat scheme in Section 5, I recover the amounts spent on car and housing insurance by income group using the 2017 French Household Budget survey from the French National Institute of Economic Studies and Statistics[10]. The sample includes 15,000 households, which I decompose in income deciles. In this survey, respondents were asked about how much they spent on many categories during year 2017, which enables me to recover the amounts spent on car and housing insurance. I use this information to compute CatNat contributions.

I denote the insurance premium π, which is the sum of the CatNat contribution π_c and the premium for other risks π_o. For housing (or car) insurance, contributions amount to 12% (or 6%) of the premium for other risks π_o. It can be summarized as follows

$$\pi_c = r\pi_o \qquad r = \{0.06, 0.12\}$$

$$\pi_c = \frac{r}{1+r}\pi$$

As I observe π in the French Household Budget survey, I can directly derive the share corresponding to the CatNat fee both for car and housing insurance.

In the survey, I only observe the premiums paid by renters and by owners of owner-occupied

[10] *Budget de Famille 2017*, INSEE. Data are confidential and must be accessed with an authorization from INSEE and through the *Centre d'Accès Sécurisé aux Données* (CASD).

homes, but I do not know the amounts paid by owners of rental, second and empty dwellings. To derive how much households pay for these types of dwellings, I run a machine learning algorithm trained on the survey data, to attribute to each observation in my sample of 29 million dwellings the estimated premium paid by owners and (if applicable) renters. Based on France Assureurs (2023), I re-scale these amounts for premiums paid by owners of second, rental and empty homes. For example, I know that premiums for second homes are on average 20% lower than premiums for owner-occupied homes. I thus re-scale all premiums for second homes accordingly.

As we use the same model for all dwellings, the first underlying assumption of this approach is that within each category of dwelling (owner-occupied, rental, second and empty homes), the criteria to decide how much to pay in insurance premium are similar. The second assumption is that the variables included in the model (dwelling's surface, number of rooms, region, resident's income and household size) are sufficient to get a reliable estimation. I argue that, at the aggregate level, most of the variation in insurance premiums should indeed be explained by these variables. To support this idea, Figure 8 compares the distribution of housing insurance premiums by income decile in the survey data and the estimated premiums on the dataset covering all dwellings in France. Panel A takes into account only premiums paid by residents of owner-occupied homes and Panel B those paid by tenants of rental homes. The correlation coefficients are respectively 96% and 97%.

3 Heterogeneous vulnerabilities

In this section, I measure the differential exposure to flood risk on various income groups by calculating the share of housing wealth that is vulnerable to these events. This ratio serves as a metric for quantifying exposure to floods in relation to total housing wealth. I have not included other forms of financial assets, since direct observations of these assets are not available in my data. My exposure index thus represents an *upper bound* estimate of the proportion of total wealth that is exposed to flooding. As I have around 1% of dwellings in my sample exposed to floods, for most households the ratio of exposed housing wealth over total housing wealth is equal to zero. In the remainder of this section, I focus on households renting or owning at least one dwelling at risk.

The level of vulnerability among tenants to flood events is particularly pronounced. Typ-

ically, their limited assets consist primarily of household furnishings contained within their rented accommodations. My sample data reveals that less than 15% of exposed tenants possess residential properties and only 28% exhibit positive financial incomes. These findings underscore that, for the majority of tenants residing in flood-prone regions, their entire wealth is highly vulnerable to the impact of flooding.

Figure 3a displays the distribution of the ratio of housing wealth at risk to total housing wealth for the four different types of dwellings[11]. I find a substantial degree of exposure among homeowners, where 100% of their housing wealth is at risk in 75% of cases. For the remaining 25% of at-risk homeowners, some may own a second property or engage in rental activities. However, less than 10% of them have less than half of their housing wealth exposed to risk. These findings underscore that the majority of homeowners at risk do not significantly diversify their real estate assets when it comes to mitigating flood-related risks. In addition, 55% of exposed homeowners have reported positive financial incomes.

The situation differs significantly for exposed landlords. Figure 3a illustrates that it is only in 5% of cases that exposed landlords have 100% of their housing wealth at risk. For 50% of exposed landlords, the proportion of their total housing wealth at risk is below 30%. Furthermore, 74% of these households report positive financial income. These findings imply that landlords exhibit a more robust risk diversification strategy: a smaller portion of their real estate assets is exposed to flood risk, and they invest more in other financial instruments.

Figure 3b illustrates that, in total, approximately 40% of exposed homeowners have their entire housing wealth exposed. This finding challenges the common perception that flood victims inevitably "lose everything." While this circumstance does apply to 40% of households, a substantial portion of exposed households has also demonstrated a greater capacity for diversification in the context of flood risk.

Finally, I regress the ratio of wealth exposed over total housing wealth against income and wealth levels among owners with at-risk properties. The results are presented in Figure 4. It becomes apparent that individuals with the lowest housing wealth levels are the most adversely affected by floods. I observe the same pattern with respect to income.

These findings underscore the substantial heterogeneity in vulnerability regarding flood

[11] As I use average house prices at the municipality level to measure the housing wealth at risk, one may argue that these prices are endogeneous and that the housing market already priced flood risk. Dubos-Paillard et al. (2023) find that the price discount of flood risk should be of 3% to 7% in France. I replicated Figure 3, but increasing the value at risk by 7% to measure an upper bound of the value at risk before the market priced flood risk. Results are very similar. Figures are available upon request.

risk. Approximately 40% of homeowners exposed to flooding have less than 40% of their total housing wealth at risk, and often possess additional financial wealth. These landlords tend to be wealthier and have higher incomes compared to the broader population. Conversely, 40% of households have their entire housing wealth exposed and display minimal diversification concerning flood risk. Furthermore, tenants, who typically own only their furnishings and lack other forms of wealth, face significant losses in the event of flooding. These observed patterns highlight the substantial variability in vulnerability, emphasizing the importance of distinguishing between these three categories of households.

4 Sorting over Flood Risk

In this section, I analyze the patterns of sorting over flood risk. I provide descriptive evidence on exposure by type of dwelling and income levels and then investigate the underlying mechanisms.

4.1 Descriptive evidence

Figure 5 shows the share of dwellings exposed to floods by type. Owner-occupied homes constitute 55% of all dwellings in France, but they make up only 40% of the dwellings in flood-exposed regions. Conversely, second homes, which account for less than 10% of all dwellings in France, are overrepresented, being twice as prevalent in flood-exposed areas. In the case of coastal flooding, second homes take up almost 50% of the exposed dwellings. Rental dwellings, while not as substantially overrepresented as second homes, still exhibit a slight overrepresentation in flood-exposed areas. These observations show that in areas prone to flooding, second homes tend to substitute for owner-occupied residences, and tenants, who are often associated with lower income levels and higher vulnerability, are not spared by flood risks.

As I also have access to the income of owners and tenants, I examine how the likelihood of residing in flood-exposed areas differs with respect to income across the different categories of dwellings. I run the following regression

$$Risk_i = \alpha + \sum_{2}^{100} \beta^k Percentile_i^k + \varepsilon_i$$

I define $Risk_i$ as an indicator variable for whether dwelling i is located in an exposed area. $Percentile_i^k$ is an indicator variable, taking a value of one if the household residing in or owning dwelling i falls within the kth percentile concerning their living standard (calculated as income divided by household size). Consequently, the parameter β^k captures the variation in the probability of being situated in an exposed area for households within the kth percentile, relative to those in the first percentile. I run this regression for three types of dwellings: owner-occupied, second homes, and rental properties. In the case of rental properties, two separate regressions are run, one where $Percentile_i^k$ represents the income of the landlord, and the other where it signifies the income of the tenant.

Results are displayed on Figure 6. Low-income homeowners exhibit a higher probability of residing in flood-exposed areas when compared to their high-income counterparts. A parallel pattern emerges when examining the flood exposure of tenants. These results collectively imply that, in the context of selecting the location for their residence, high-income households are underrepresented in flood-prone areas as compared to their low-income counterparts.

As we discerned from Figure 5, there is a notable overrepresentation of second homes in flood-prone areas. Figure 6 provides a deeper understanding of this phenomenon. Households situated within the middle of the income distribution exhibit a higher degree of overrepresentation in these areas. Furthermore, the degree of overrepresentation at the upper end of the income spectrum is akin to that at the lower end. This pattern suggests that as income increases, households display a preference for locating their second homes along coastal regions, which happen to be flood-prone areas. However, once income surpasses a certain threshold, around the 80th percentile in my context, households appear to shift their preference, choosing to own second homes in alternative locations. In France, high-income households frequently opt for second homes in mountainous regions or in the Paris area (Châtel et al., 2021). This may explain the U-shaped curve that emerges.

To get more sense about the size of these effects, I compute the income gap between exposed and non-exposed areas. I run the following regression:

$$Log(Income_i) = \alpha + \beta Risk_i + \varepsilon_i$$

Table 1 displays the beta coefficients.

Looking at Panel A and focusing exclusively on homeowners (column 1), it appears that those situated in flood-exposed areas exhibit, on average, a lower income by approximately 6%

when compared to the rest of the population. Knowing that homeowners are underrepresented in flood-prone areas from Figure 5, it appears that this underexposure is primarily driven by the decision of high-income households to avoid residing in areas characterized by flood risk, resulting in an income gap of 6% between safe and exposed areas.

For second homes (column 2), we observe an income gap of 6% that is due to the preference for other amenities than the coast for high-income owners of second homes. However, all income groups remain over-represented in flood-risk areas as documented in Figure 5.

In the case of landlords who own rental properties (column 3), a similar income gap of 6% is observed. When examining Figure 5, it seems that the proportion of rental properties remains consistent between flood-exposed and non-exposed areas. This implies that within the category of rental properties, high-income landlords tend to exhibit a propensity to steer clear of flood-prone regions, while landlords with lower and middle incomes are more inclined to possess rental properties in areas susceptible to flooding. These two effects effectively balance each other, resulting in the overall share of rental properties in flood-exposed areas being equivalent to that in non-exposed areas. However, the composition of rental properties in flood-exposed areas predominantly comprises households with lower incomes. This scenario suggests that high-income landlords diversify better their real estate portfolios concerning flood risk.

The income gap for empty dwellings (column 4) is small and almost not significant.

I find that low-income tenants (column 5) exhibit a pronounced inclination to settle in flood-risk areas. This sorting behavior leads to a substantial income gap of 9% between flood-exposed and safe areas. Low-income tenants are typically representative of households with limited financial resources. These findings underscore the fact that, in addition to their high vulnerability to floods due to low diversification, low-income tenants also confront an elevated level of exposure to these risks.

In Panel B, I add successively the different types of dwellings to see how they contribute to the total income gap between exposed and non-exposed areas. In column 1, we note the previously identified income gap of 6% for homeowners. However, when second homes are introduced into the equation, this gap nearly diminishes and approaches insignificance, nearly equaling zero. These results collectively suggest that the 6% income gap observed for homeowners is offset by the substantial overexposure of second homes. This finding suggests a noteworthy substitution effect, primarily driven by high-income households. They tend to shift from owner-occupied residences to second homes in flood-exposed areas, but this behavior is not mirrored by low-income households. When rental and empty dwellings are included in the

analysis, the income gap increases marginally, but the overall aggregate gap remains relatively modest. In conclusion, owners' income is only 0.5% smaller in flood-exposed areas compared to safe ones.

The overall picture that emerges suggests that both high- and low-income homeowners face exposure to flood risks in similar proportions. However, the impact of this exposure weighs far more heavily on low-income homeowners. Their exposure tends to occur through their primary place of residence, as opposed to second homes, which are typically used for only a few weeks each year. This distinction highlights that despite a marginal income gap among homeowners at the aggregate level, low-income homeowners encounter a considerably higher level of vulnerability in comparison to their high-income counterparts.

4.2 Mechanisms

4.2.1 Within-municipality variation

As I have access to very fine-grain data, I reproduce the same analysis as in Section 4.1 but looking at within-municipality variation. I run the following regression

$$Risk_i = \alpha + \sum_{2}^{100} \beta^k Percentile_i^k + \gamma_m + \varepsilon_i$$

I incorporate municipality fixed effects (γ_m) to control for local variations. It's important to note that only municipalities that have at least one dwelling located in exposed areas are included in the analysis. The coefficient β quantifies the variation in the probability of exposure to flood risk in comparison to the first income percentile within the same municipality. On average, there are approximately 1,000 dwellings within each municipality. Therefore, the β coefficient serves as a measure of the probability of being situated in a flood-prone area within a relatively small pool of approximately 1,000 dwellings, on average. The findings from this analysis are presented in Figure 7.

I also reproduce Table 1 adding municipality fixed effects. I run the following regression

$$Log(Income_i) = \alpha + \beta Risk_i + \gamma_m + \varepsilon_i$$

Results are displayed on Table 1.

Under these model specifications, the observed effects should not be driven by variations in

amenity preferences between municipalities. Instead, these effects could potentially be attributed to two distinct channels:

- *Channel 1*: preferences related to amenities within municipalities. For instance, households may exhibit a penchant for features such as a direct view of the coast, thereby guiding their residential choices.
- *Channel 2*: strategic location behaviors that aim to circumvent exposure to flood risk. Households may adopt a calculated approach to their residential choices in order to mitigate the risk of being located in flood-prone areas.

At present, distinguishing between these two channels is not straightforward. However, Iplan tostudy the housing market composition at the boundary of the flood map to see if there are strategic behavior of flood risk avoidance for different types of dwellings.

In Figure 7, we observe the same patterns as before for owner-occupied homes and tenants. Looking at Panel A of Table 2, we see that the income gap is of similar magnitude for homeowners (column 1) and divided by two for tenants (column 5). These results suggest that when it comes to choosing their place of living, households in the top of the income distribution sort less over flood risk areas. It seems unlikely that high-income households enjoy less the view of the coast than low-income ones (channel 1), which suggests that these households are more likely to take flood risk into account when deciding on their place of living (channel 2). For second homes and owners of rental dwellings (columns 2 and 3), the previously noted patterns appear to dissipate, and the income gap becomes statistically insignificant. It suggests that the patterns observed earlier were primarily influenced by disparities in amenities between municipalities, rather than being driven by deliberate flood risk avoidance decisions or the influence of within-municipality amenities.

Finally, looking at Panel B of Table 2, we see that the inclusion of second homes results in a substantial reduction of the income gap between owners (columns 1 and 2). This outcome illustrates that, even at the local level, there exists a significant concentration of second homes within flood-exposed areas. The addition of rental and empty dwellings into the analysis (columns 3 and 4) yields only a marginal effect on the income gap for owners between flood-exposed and non-exposed areas.

4.2.2 Avoidance behavior - work in progress

As I have access to the floor number at which every household resides, the objective of this section will be to see whether high-income households also avoid exposure to floods by locating their dwellings on upper floors.

4.2.3 Heterogeneous updating processes - work in progress

I plan to use delineation maps of past flood events to see whether income groups update differently to floods. The objective will be to see whether after a flood event, high-income households leave flooded areas and prefer to have establish their second home there. This could explain the patterns we observed earlier on.

5 Implications for policy

In this section, I discuss the consequences of these findings on two adaptation policies: home buyout and subsidized flood insurance.

5.1 Home buyout programs

Home buyout programs involve governments purchasing at-risk properties, often at a negotiated price, for demolition. In France, the *Fonds Barnier* manages such initiatives, especially after major floods. In the United States, the Federal Emergency Management Agency oversees a buyout program for addressing similar risks, focusing on purchasing and removing vulnerable properties.

Home buyout programs offer a "fair" solution to assist victims of natural disasters. These programs enable affected individuals to sell their properties at a fair price, facilitating a move from high-risk areas to safer locations. Moreover, it can be economically beneficial, especially when a dwelling is subjected to recurrent flooding, as purchasing and demolishing such properties often becomes a cost-effective choice.

Home buyout programs, while effective, face significant challenges due to their high costs and logistical complexities. Many households in flood-prone areas are willing to sell their properties at market prices to entities like the *Fonds Barnier* and relocate. However, limited resources often prevent these programs from meeting the full extent of the demand they receive.

Typically, home buyout policies are implemented after specific events, and they often lack a targeted approach that considers households' vulnerabilities. In the subsequent section, I conduct a simulation of a policy that involves purchasing all at-risk homes in France. While this scenario may be highly impractical, it serves as a useful exercise in comprehending how improved targeting in home buyout policies can significantly reduce their overall costs.

The results, as displayed in Table 3, are presented in the form of the share of the total buyout cost for the proposed policy. This table effectively provides the estimated overall cost of an ambitious buyout policy, contingent on the government's decision to target specific categories of households. For instance, if the government opts to purchase all at-risk owner-occupied homes belonging to the bottom 50%, the cost would only amount to 18% of the total buyout expense. These findings underscore the potential for significantly reducing the cost of buyout policies through more precise targeting strategies.

5.2 Subsidized weather insurance

As discussed in Section 1, the French CatNat insurance system against natural disasters provides coverage to 98% of households in the country. The financing of this system relies on an additional premium of 12% for housing insurance contracts and 6% for automotive insurance contracts. The key advantage of this system lies in its ability to offer extensive coverage at an affordable cost to nearly all households in France, a goal that many other countries such as the United States often fail to achieve (Wagner, 2022). However, this system operates through cross-subsidies from safe to exposed areas, giving rise to distributive implications. In this section, we delve into these distributive considerations.

Using data from the French Household Budget survey presented in Section 2.3, I assess the weight of CatNat premiums on households' budget. Panel A of Figure 9 displays the total amount paid in CatNat additional premium across income deciles, following the methodology described in Section 2.3. Panel B shows the ratio of these additional premiums over disposable income. We see that while higher income households pay contribute more to the system in total, CatNat contributions impose a significantly heavier burden on households at the bottom 10%, four times higher than for those at the top 10%. CatNat premiums as a share of income are higher for all income deciles in comparison to the top 10%, displaying a consistent diminishing trend with respect to income. These findings illuminate the regressive nature of these additional premiums.

Furthermore, considering the findings from Section 4, it appears that the CatNat system lacks efficient targeting of households with the greatest needs. Despite the fact that both high- and low-income homeowners face exposure to natural disasters to a similar extent, the underlying dynamics differ significantly. Low-income homeowners experience heightened vulnerability through their primary residences, while high-income individuals are more exposed through their second homes. In this context, the CatNat system extends its benefits uniformly to all income groups, regardless of the substantial disparities in vulnerability.

To gain a deeper understanding of how the CatNat system redistributes money across income groups, I calculate the distribution of *CatNat premiums* and compare it to the distribution of *expected damages*. CatNat premiums represent the premiums displayed in Figure 9. Expected damages are the anticipated compensation that a household would receive if their at-risk dwelling were to be flooded. I estimate these damages based on the dwelling's value[12]. For each decile d I measure

$$Catnat_Premiums_d = \frac{\sum_{i \in I_d} CatNat_premium_i}{\sum_i CatNat_premium_i}$$

with $CatNat_premium_i$ being the CatNat premium paid by household i and I_d being households in decile d.

$$Expected_Damages_d = \frac{\sum_{i \in I_d} \sum_{j \in J_i} Risk_j * Ground_floor_j * Value_j}{\sum_i \sum_{j \in J_i} Risk_j * Ground_floor_j * Value_j}$$

with $Risk_j$ being an indicator variable for whether dwelling j is located in a risky area, $Ground_floor_j$ an indicator variable for whether dwelling j is is on the ground floor and $Value_j$ being dwelling j's value based on its surface and the average price per meter squared in the municipality. J_i is the universe of dwellings owned by household i.

Panel A of Figure 10 illustrates the distribution of *CatNat premiums* and *expected damages* across income deciles. It appears that the first decile pays 7.5% of the total CatNat premiums and bears 6% of the total expected damages. Consequently, assuming that total premiums equal total damages, the first decile incurs a net loss from the system, given that the expected damages are lower than the premiums they pay. In contrast, the top decile receives more compensation

[12]Since the value of the dwelling can capture characteristics beyond the core value of the building, I conducted a parallel analysis using the surface area of the dwellings and obtained very similar results. Detailed results are available upon request.

from CatNat than what they pay in premiums.

Drawing from the estimates in Panel A, Panel B of Figure 10 shows the average net gains or losses for each income decile for a total amount of damages of 1 billion euros, which approximately corresponds to the average annual flood-related losses in France (France Assureurs, 2021a). When summing the net losses of the bottom 50% (or equivalently the gains of the top 50%), we observe a net transfer of approximately 50 million euros, which represents about 5% of total annual losses. This amount is redistributed from the bottom 50% to the top 50%.

It's important to acknowledge that this simulation is a simplified version of the CatNat system. Firstly, the system covers not only flood-related risks but also those related to subsidence caused by droughts, storms, and hurricanes. However, in this representation, we simplify the CatNat system as if it exclusively covered floods. Secondly, approximately 20% of the expected damages are directed toward companies on average. For the purposes of this analysis, we focus solely on residential dwellings and abstract from these commercial claims. Thirdly, in practice, all at-risk buildings are not flooded simultaneously. What we are measuring here are the "expected" damages, which represent a theoretical estimate of average damages over time. Lastly, we simplify the scenario by excluding the possibility of the CatNat system operating at a deficit. In the real world, the system might face situations where the amount of money collected through premiums does not precisely match what is required to compensate households affected by floods.

The simplifications made in this analysis imply that the results should not be regarded as a comprehensive assessment of the CatNat system. Instead, they provide an estimate of the magnitude of redistribution in favor of flood-prone areas because of a public system like CatNat.

Despite enabling all households to be covered at an affordable price, the CatNat system is thus financed through regressive premiums that participate to subsidize the second-homes at risk of high-income landlords. The system could be enhanced by implementing a more progressive premium structure. This could involve indexing premiums to the price per square meter or introducing an additional premium specifically for second homes. These adjustments would serve to create a more equitable and responsive system that better aligns with the varying needs and vulnerabilities of households.

6 Conclusion

Most of the literature studying the incidence of flooding have primarily relied on aggregate data. This paper leverages a novel dataset that enables to study flood exposure at the most granular level possible. I cover 80% of dwellings in France, furnishing comprehensive information regarding the income and housing wealth characteristics of both property owners and tenants.

Specifically, low-income tenants and homeowners often have their entire wealth, up to 100%, exposed to flooding. In contrast, a large proportion of high-income individuals and affluent landlords, approximately 50% of them, have less than one-third of their total wealth exposed to flood risk. This finding underscores the substantial heterogeneity in vulnerability across different income and wealth categories.

Second, low-income homeowners and tenants tend to more frequently choose to reside in high-risk areas compared to their high-income counterparts. This leads to a notable income gap between exposed and non-exposed areas, amounting to 6% for homeowners and 9% for tenants.

Third, middle- to high-income households to prefer situating their second homes in high-risk areas. Consequently, there is a considerable over-representation of second homes in these risk-prone regions, amounting to a factor of two.

Fourth, the pronounced over-exposure of second homes in high-risk areas is so significant that it effectively neutralizes the 6% income gap between exposed and non-exposed property owners at the aggregate level. Consequently, this phenomenon leads to a comparable distribution of assets across income groups. However, it's important to note that this equitable distribution hides the heightened vulnerability of low-income property owners as compared to high-income owners of second-homes.

Fifth, these patterns appear to be influenced, at least in part, by high-income households actively avoiding the placement of their primary residences in flood-prone areas, rather than solely being driven by amenity preferences. On the contrary, the location of second homes seem to be driven by differences in amenities across municipalities. However, a more detailed examination is warranted to effectively disentangle the contributions of these factors.

These findings carry significant implications for adaptation policies. By targeting beneficiaries more effectively, the cost of buyback policies can be substantially reduced. Furthermore, subsidized insurance schemes like the French CatNat system could be enhanced by set-

ting premiums that do not burden low-income households and by focusing on households in greater need, rather than channeling resources to repair second homes used only a few weeks per year. Given the growing popularity of such insurance schemes, especially in the United States where the issue of under-coverage against natural disaster risks is pressing (Wagner, 2022), these findings offer valuable insights for policymakers seeking to design similar systems. Potential strategies include indexing premiums to the price per square meter or implementing an additional fee for second homes, both of which can make the system more progressive and prioritize those in need of coverage.

References

André, M. and O. Meslin (2021). Et pour quelques appartements de plus: Étude de la propriété immobilière des ménages et du profil redistributif de la taxe foncière. *Insee, Document de travail* (2021-04).

Bach, L., L. E. Calvet, and P. Sodini (2020). Rich pickings? risk, return, and skill in household wealth. *American Economic Review 110*(9), 2703–2747.

Bakkensen, L. A. and L. Ma (2020). Sorting over flood risk and implications for policy reform. *Journal of Environmental Economics and Management 104*, 102362.

Balboni, C. A. (2019). *In harm's way? infrastructure investments and the persistence of coastal cities*. Ph. D. thesis, London School of Economics and Political Science.

Banzhaf, S., L. Ma, and C. Timmins (2019). Environmental justice: The economics of race, place, and pollution. *Journal of Economic Perspectives 33*(1), 185–208.

Ben-Shahar, O. and K. D. Logue (2016). The perverse effects of subsidized weather insurance. *Stan. L. Rev. 68*, 571.

Bin, O., J. Bishop, and C. Kousky (2017). Does the national flood insurance program have redistributional effects? *The BE Journal of Economic Analysis & Policy 17*(4).

CCR (2022). Les catastrophes naturelles en france - bilan 1982-2022. *Caisse Centrale de Réassurance*.

CCR (2023). Etude sur les consequences du changement climatique sur le coût des catastrophes naturelles en france à horizon 2050. Technical report.

Chancel, L., T. Piketty, E. Saez, and G. Zucman (2022). *World inequality report 2022*. Harvard University Press.

Charpentier, A., L. Barry, and M. R. James (2022). Insurance against natural catastrophes: balancing actuarial fairness and social solidarity. *The Geneva Papers on Risk and Insurance-Issues and Practice 47*(1), 50–78.

Châtel, F., N. Cochez, and M.-P. de Bellefon (2021). Deux résidences secondaires sur trois sont détenues par un ménage de 60 ans ou plus. *Insee Première 1871*.

Commission, E. (2022). Decision (eu) 2022/591 of the european parliament and of the council of 6 april 2022 on a general union environment action programme to 2030.

Cour des comptes (2022). Sols argileux et catastrophes naturelles. Technical report.

Dinan, T., P. Beider, and D. Wylie (2019). The national flood insurance program: Is it financially sound? *Risk Management and Insurance Review 22*(1), 15–38.

Dubos-Paillard, E., E. Lavaine, and K. Millock (2023). Flood risk information: Evidence from paris housing markets. *AFSE Working Paper*.

Fagereng, A., L. Guiso, D. Malacrino, and L. Pistaferri (2020). Heterogeneity and persistence in returns to wealth. *Econometrica 88*(1), 115–170.

France Assureurs (2021a). Impact du changement climatique sur l'assurance à l'horizon 2050. Technical report.

France Assureurs (2021b). L'assurance des évènements naturels en 2021. Technical report.

France Assureurs (2023). L'ssurance habitation en 2022. Technical report.

Grislain-Letrémy, C. (2018). Natural disasters: exposure and underinsurance. *Annals of Economics and Statistics/Annales d'Économie et de Statistique* (129), 53–83.

Grislain-Letrémy, C. and C. Peinturier (2010). Le régime d'assurance des catastrophes naturelles en france.

Hallegatte, S., C. Green, R. J. Nicholls, and J. Corfee-Morlot (2013). Future flood losses in major coastal cities. *Nature climate change 3*(9), 802–806.

Holladay, J. S. and J. A. Schwartz (2010). Flooding the market: the distributional consequences of the nfip. *Institute for Policy Integrity, Policy Brief 7*.

Hudson, P. (2018). A comparison of definitions of affordability for flood risk adaption measures: a case study of current and future risk-based flood insurance premiums in europe. *Mitigation and adaptation strategies for global change 23*(7), 1019–1038.

INSEE (2023). Fichier localisé social et fiscal (filosofi) - sources et méthodes. Technical report.

Jaffee, D. M. and T. Russell (1997). Catastrophe insurance, capital markets, and uninsurable risks. *Journal of Risk and Insurance*, 205–230.

Kocornik-Mina, A., T. K. McDermott, G. Michaels, and F. Rauch (2020). Flooded cities. *American Economic Journal: Applied Economics 12*(2), 35–66.

Kousky, C. and H. Kunreuther (2014). Addressing affordability in the national flood insurance program. *Journal of Extreme Events 1*(01), 1450001.

Masters, J. (2022). Third-costliest year on record for weather disasters in 2021: $343 billion in damages. Technical report.

McAneney, J., D. McAneney, R. Musulin, G. Walker, and R. Crompton (2016). Government-sponsored natural disaster insurance pools: A view from down-under. *International Journal of Disaster Risk Reduction 15*, 1–9.

Mohai, P., D. Pellow, and J. T. Roberts (2009). Environmental justice. *Annual review of environment and resources 34*, 405–430.

Osberghaus, D. (2021). Poorly adapted but nothing to lose? a study on the flood risk–income relationship with a focus on low-income households. *Climate Risk Management 31*, 100268.

Rözer, V. and S. Surminski (2021). Current and future flood risk of new build homes across different socio-economic neighbourhoods in england and wales. *Environmental Research Letters 16*(5), 054021.

Sansévérino-Godfrin, V. (1996). *L'État, les compagnies d'assurance et les risques majeurs*. Ph. D. thesis, Nice.

Sastry, P. (2022). Who bears flood risk? evidence from mortgage markets in florida. *Job Market Paper*.

Sayers, P., E. C. Penning-Rowsell, and M. Horritt (2018). Flood vulnerability, risk, and social disadvantage: current and future patterns in the uk. *Regional environmental change 18*(2), 339–352.

Smith, A. B. (2021). 2020 us billion-dollar weather and climate disasters—in historical context. In *101st American Meteorological Society Annual Meeting*. AMS.

Smith, A. B. and R. W. Katz (2013). Us billion-dollar weather and climate disasters: data sources, trends, accuracy and biases. *Natural hazards 67*(2), 387–410.

Taylor, D. E. (2000). The rise of the environmental justice paradigm: Injustice framing and the social construction of environmental discourses. *American behavioral scientist 43*(4), 508–580.

Wagner, K. R. (2022). Designing insurance for climate change. *Nature Climate Change*, 1–3.

Walker, G. P. et al. (2006). *Addressing environmental inequalities: flood risk*. Bristol: Environment Agency.

Wing, O. E., W. Lehman, P. D. Bates, C. C. Sampson, N. Quinn, A. M. Smith, J. C. Neal, J. R. Porter, and C. Kousky (2022). Inequitable patterns of us flood risk in the anthropocene. *Nature Climate Change 12*(2), 156–162.

7 Figures and Tables

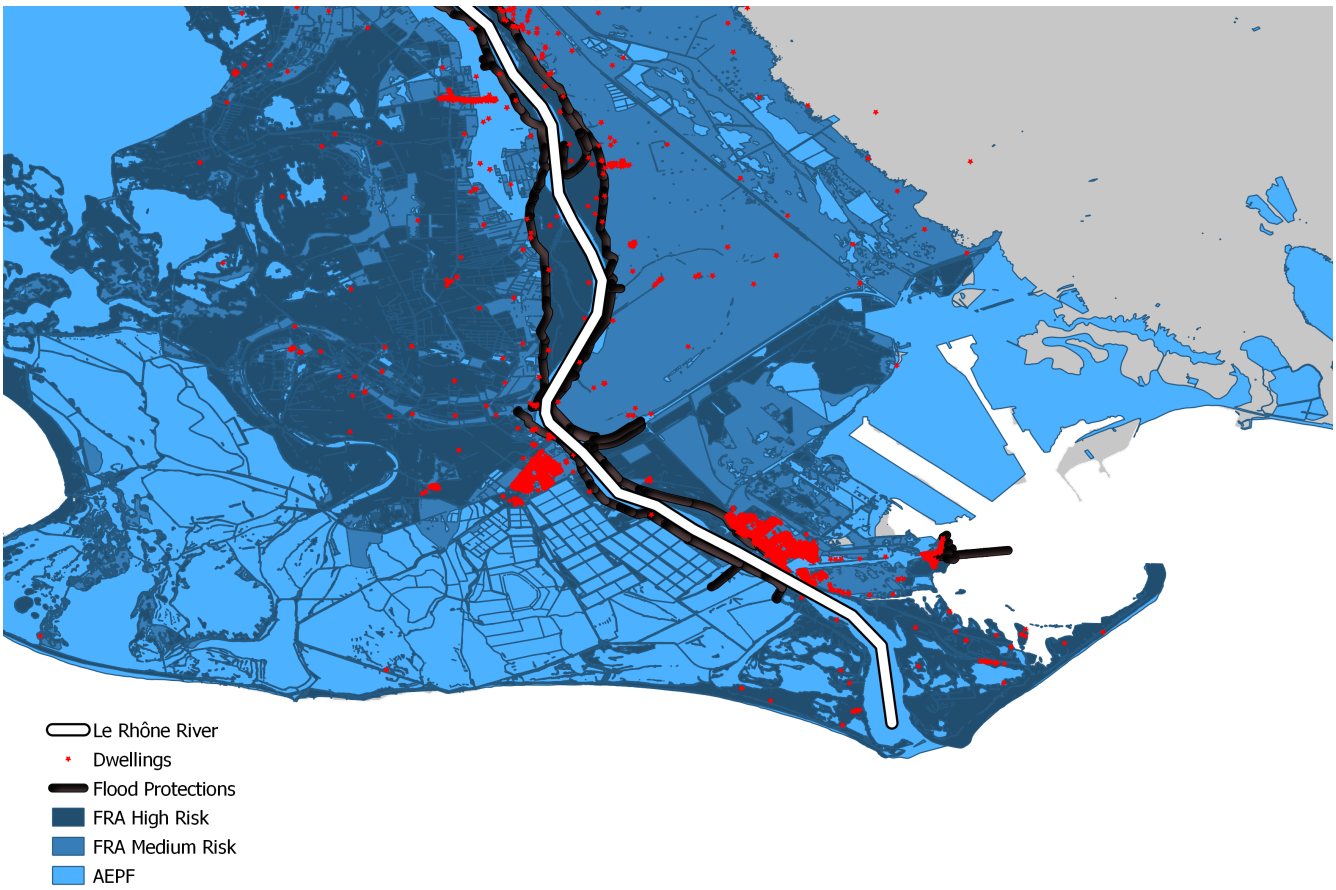
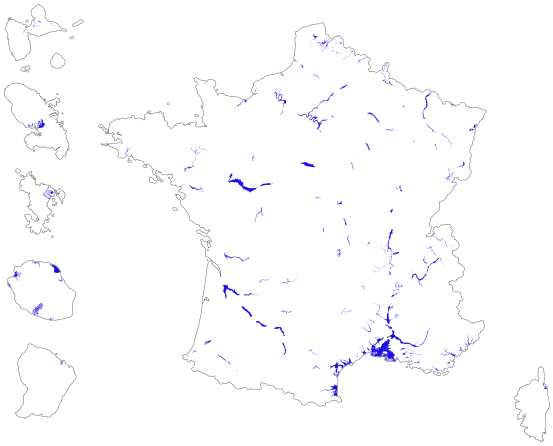
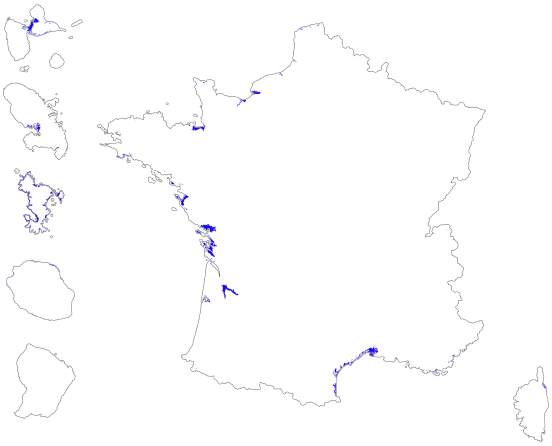


Figure 1: Coverage of FRAs and AEPF in Port-Saint-Louis-du-Rhône

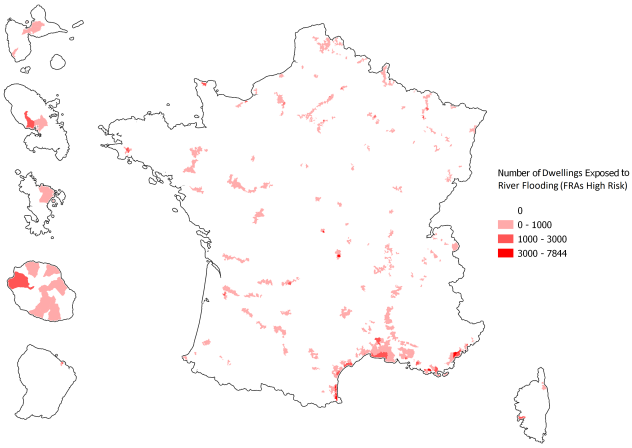
Notes. Every red dot corresponds to the location of a dwelling. The grey area is a part of France that is not affected neither according to FRAs nor by AEPF maps. The river Le Rhône is represented in white. In dark blue is the FRAs High Risk zoning. In mid-blue the FRAs Medium Risk zoning. The AEPF map is represented in light blue. Flood protections are displayed in black.



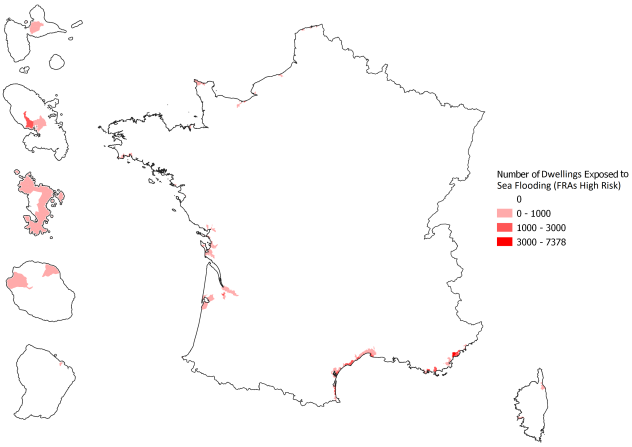
(a) Coverage of FRAs river flooding



(b) Coverage of FRAs coastal flooding



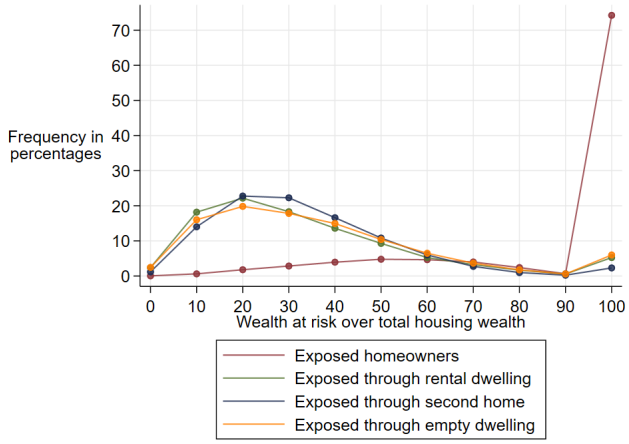
(c) Exposure to FRAs river flooding



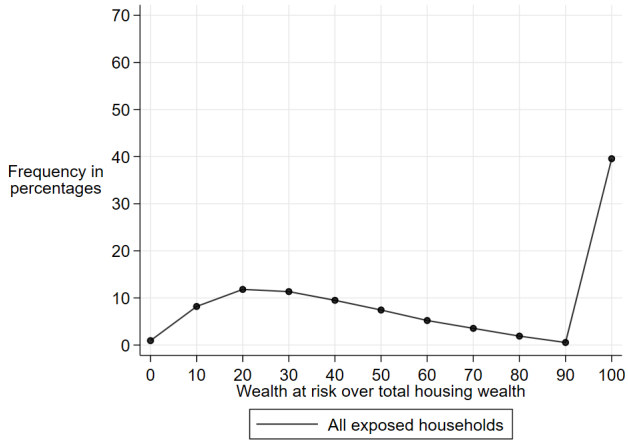
(d) Exposure to FRAs coastal flooding

Figure 2: FRA Maps of Exposure

Notes. On the center of each figure is Metropolitan France and on the left hand side are the five overseas departments. From the top to the bottom: La Guadeloupe, La Martinique, Mayotte, La Réunion and La Guyane. In Figures 2a and 2b, the blue zones are areas exposed to flooding according to FRAs. Figures 2c and 2d display the number of dwellings exposed to high risk by municipality.



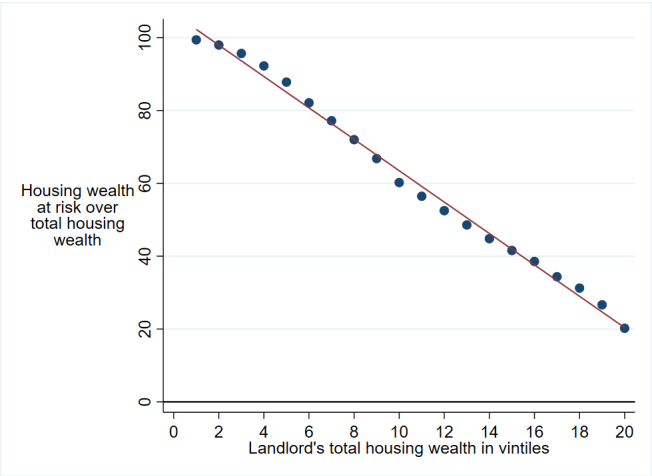
(a) By type of dwelling



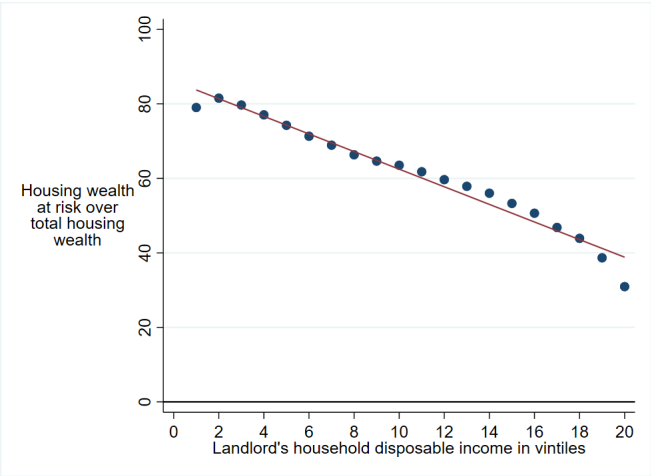
(b) Aggregate distribution

Figure 3: Distribution of the ratio of housing wealth at risk over total housing wealth for exposed owners

Notes. The Figure displays the distribution of the ratio of housing wealth at risk over total housing wealth for all exposed owners. Panel (a) differentiates by type of dwelling and Panel (b) gathers all owners together.



(a) By housing wealth vintiles



(b) By income vintiles

Figure 4: Ratio of housing wealth at risk over total housing wealth regressed on wealth and income levels

Notes. Panel (a) shows the distribution of the ratio by vintiles of housing wealth and Panel (b) by income vintiles. The red line is the linear fit.

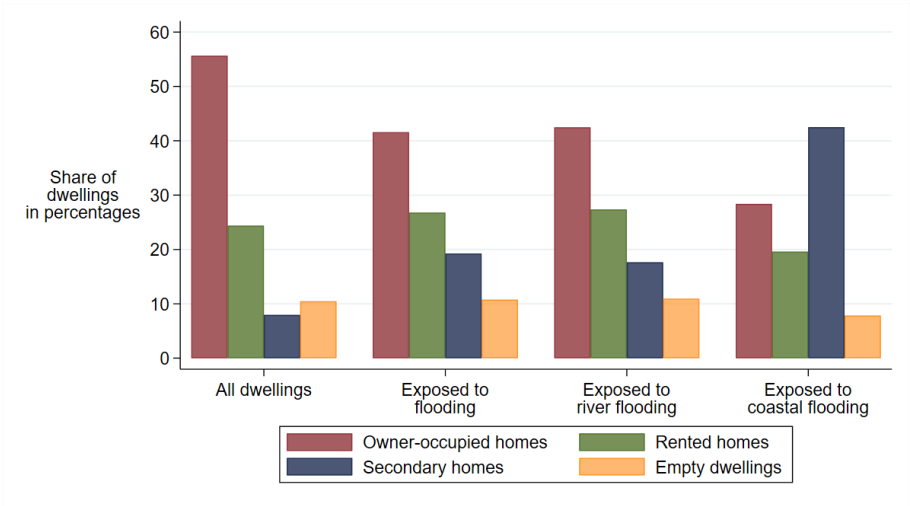


Figure 5: Exposure to flood risk by type of dwelling

Notes. The Figure displays the share of each dwelling category at the national scale, in flood exposed areas, in areas exposed to river flooding and in areas exposed to coastal flooding.

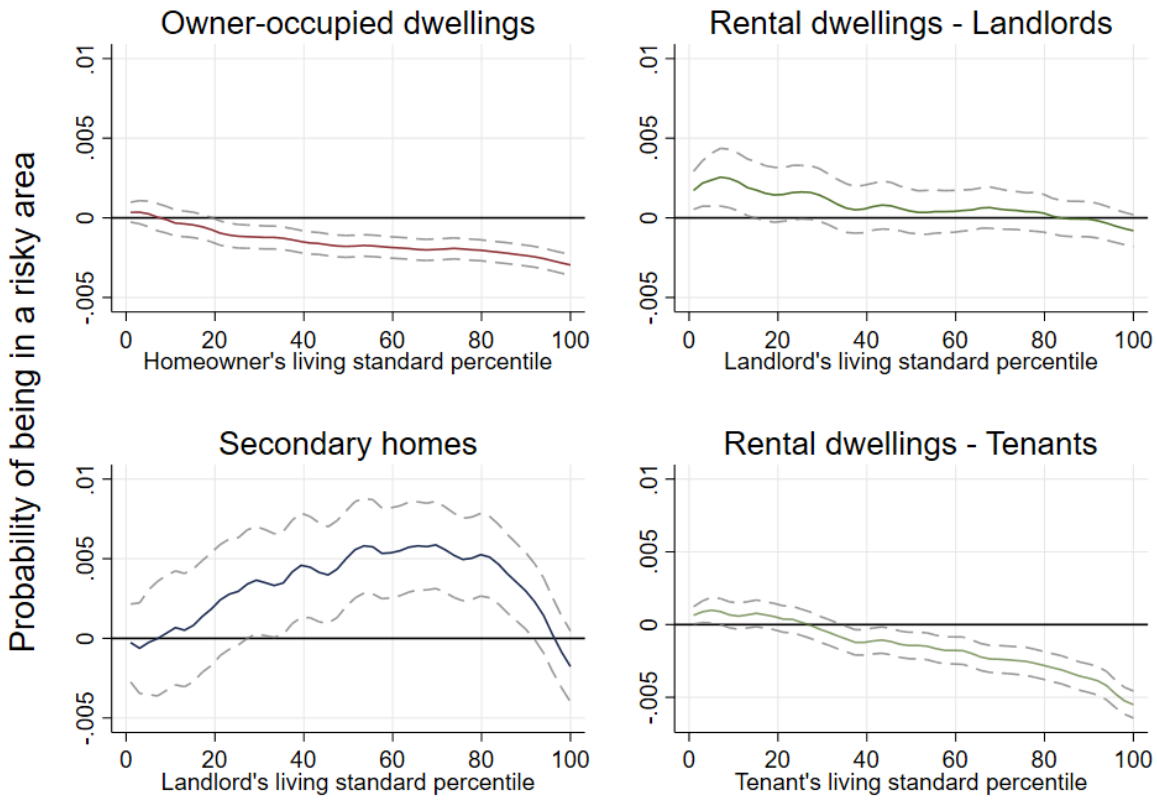


Figure 6: Probability of being exposed relative to the first income percentile by dwelling type

Notes. The Figure plots the beta coefficient of the regression $Risk_i = \alpha + \sum_2^{100} \beta^k Percentile_i^k + \varepsilon_i$ detailed in Section 4.1 for owner-occupied, rental and second homes. The regression line are smoothed using kernel-weighted local polynomial smoothing. Percentile 1 is omitted on the graph and is the reference value.

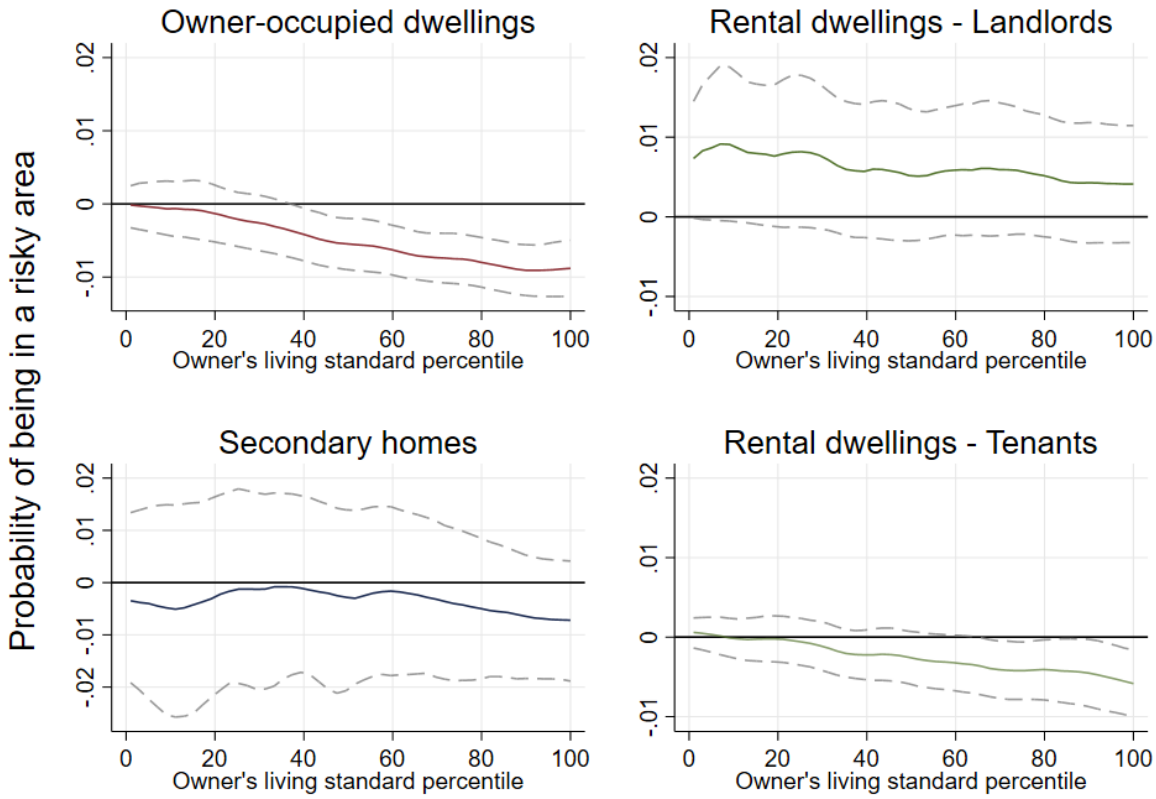
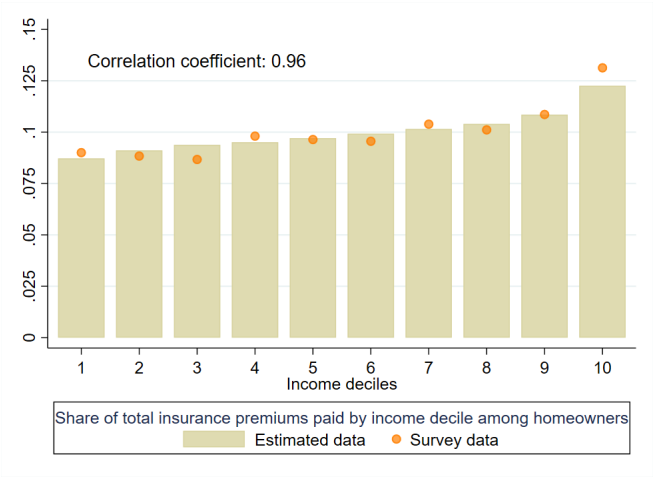
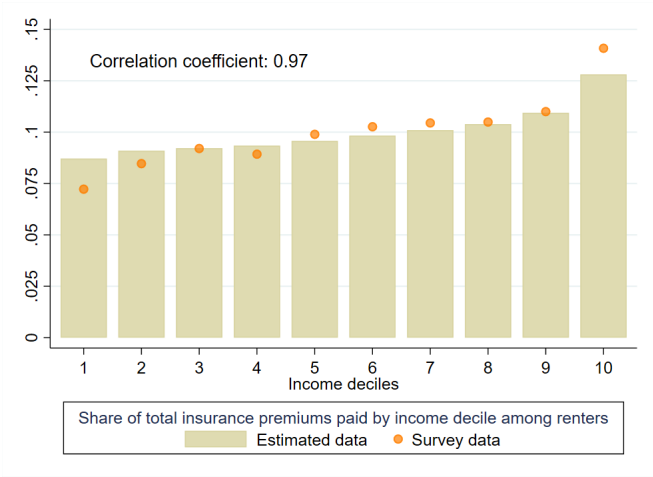


Figure 7: Within-municipality probability of being exposed relative to the first income percentile by dwelling type

Notes. The Figure plots the beta coefficient of the regression $Risk_i = \alpha + \sum_2^{100} \beta^k Percentile_i^k + \gamma_m + \varepsilon_i$ detailed in Section 4.1 for owner-occupied, rental and second homes. γ_m correspond to municipality fixed effects. Only municipalities with at least one dwelling exposed are included in the regression. The regression line are smoothed using kernel-weighted local polynomial smoothing. Percentile 1 is omitted on the graph and is the reference value.



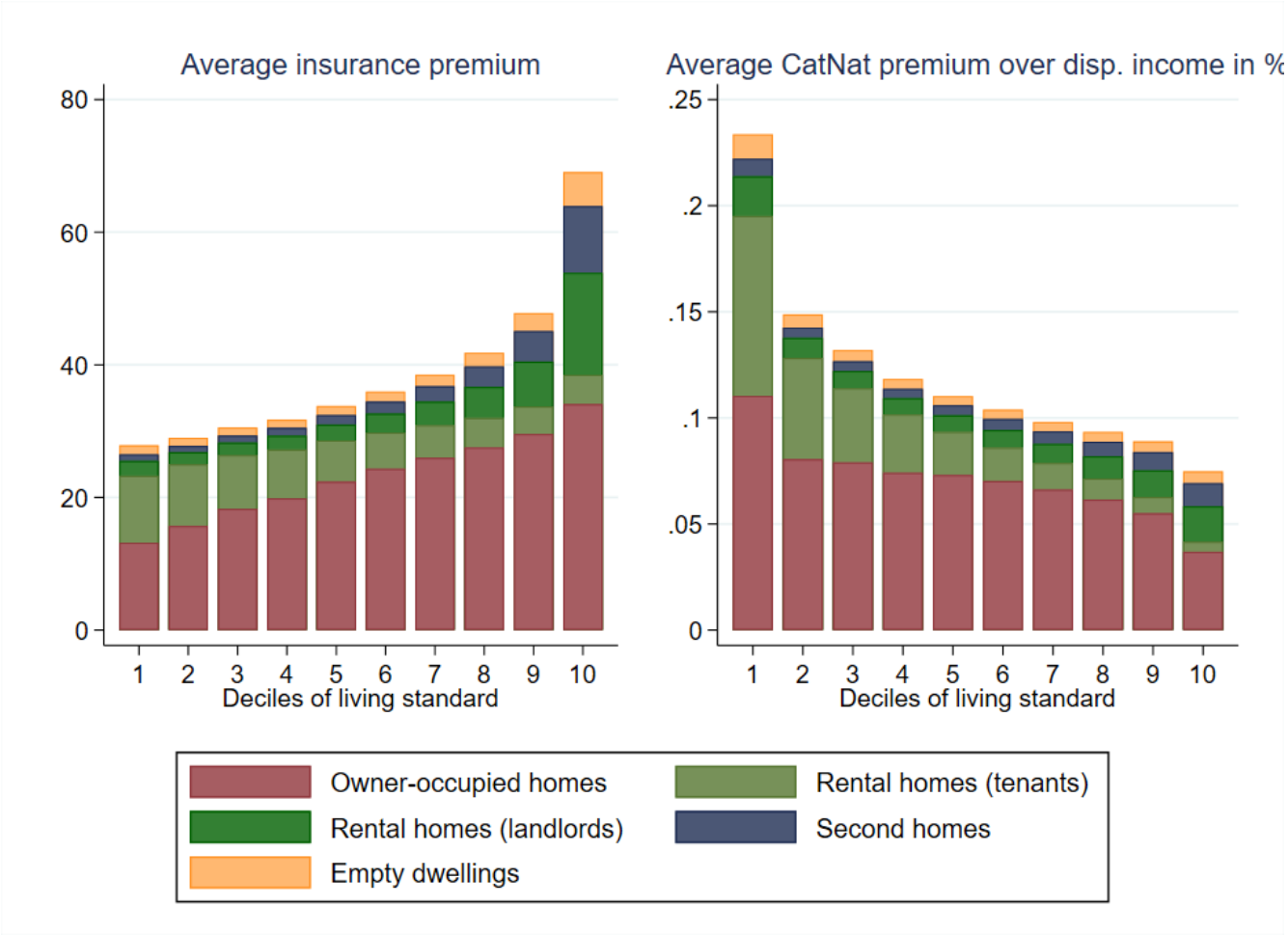
(a) For homeowners



(b) For tenants

Figure 8: Comparison between predicted home insurance premiums and survey data

Notes. The survey data gives the amount paid by homeowners and tenants in home insurance. Based on this information and a Machine Learning algorithm described in Section 2.3, I derive insurance premiums at the household level for all dwellings in France. Figures 8a and 8b compare predicted with actual survey data by income decile for respectively homeowners and tenants. Interpretation: homeowners in the bottom 10% pay 8.9% of the total amount of insurance premiums in France according to survey data (orange dot), and the predicted value (yellow bar) is 8.7%.

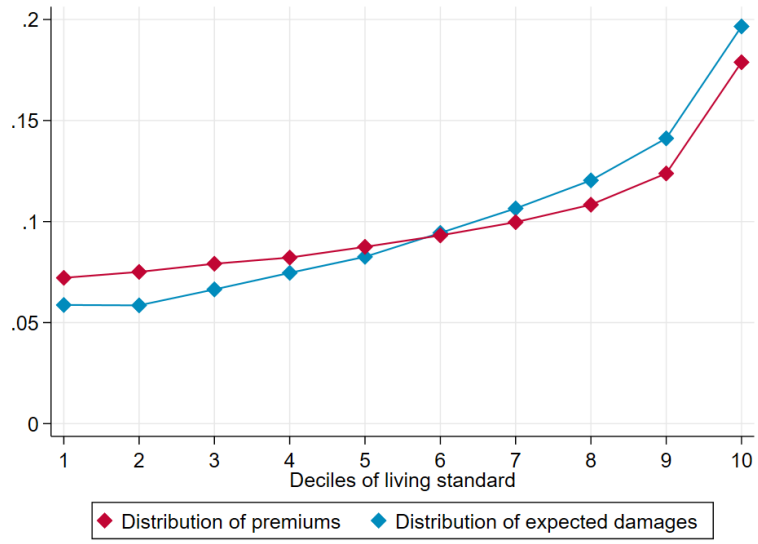


(a) Average CatNat premium

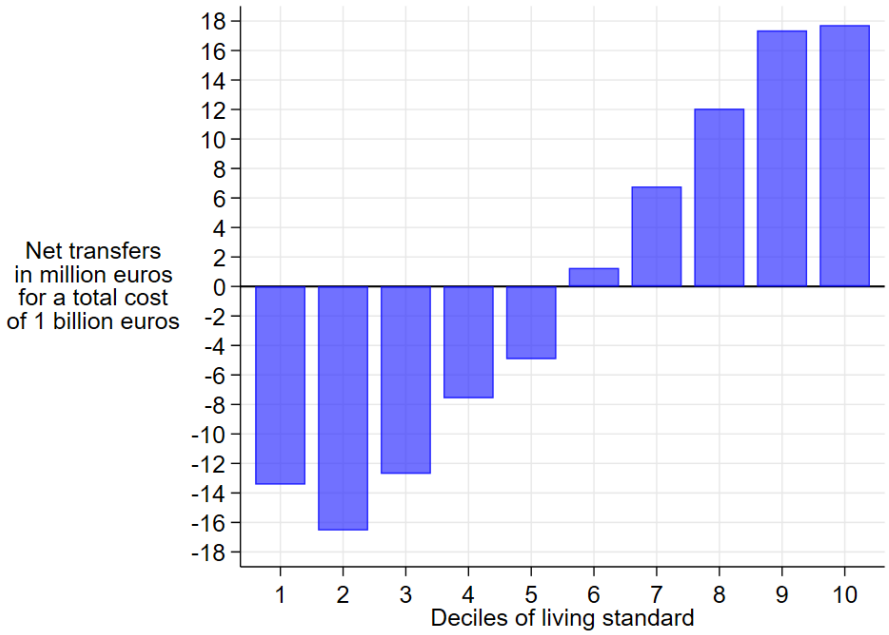
(b) As a share of disposable income

Figure 9: Distribution of CatNat premiums by income decile

Notes. CatNat insurance premiums are obtained following the Machine Learning prediction and the formula described in Section 2.3. There are five categories of households paying home insurance CatNat premiums: homeowners, tenants, owners of rental homes, owners of second homes and owners of empty dwellings. In Panel A, I sum the total amount of premiums by category and divide these amounts by the number of households in each decile to get the average premium by income decile. In Panel B, I divide these amounts by average disposable income.



(a) CatNat premiums and expected damages distributions



(b) Net expected transfers for a total loss of 1 billion euros

Figure 10: Net expected transfers induced by the CatNat system

Notes. Panel A plots the distribution of CatNat premiums and expected damages by income decile. E.g., the bottom 10% pays around 7.5% of CatNat premiums and bears 6% of expected damages. Panel B is obtained by computing the gap between the two distributions and multiplying it by 1 billion. It displays the net losses or gains by income decile.

Panel A: Income gap by type of dwelling

	Owner's log living standard				Tenant's log living standard
	(1)	(2)	(3)	(4)	(5)
Dwelling located in a risky area	-0.058***	-0.057***	-0.057***	-0.012*	-0.088***
	(0.002)	(0.002)	(0.003)	(0.005)	(0.003)
Owner-occupied dwellings	Yes	No	No	No	No
Second homes	No	Yes	No	No	No
Rental dwellings	No	No	Yes	No	Yes
Empty dwellings	No	No	No	Yes	No
Observations	16309325	2942577	7100906	3047349	7100906

*Signif. Codes: ***: 0.001, **:0.01, *:0.05*
Robust standard errors in parentheses

Panel B: Aggregate income gap

	Log living standard			
	(1)	(2)	(3)	(4)
Dwelling located in a risky area	-0.058***	0.003	-0.005***	-0.005***
	(0.002)	(0.001)	(0.001)	(0.001)
Owner-occupied dwellings	Yes	Yes	Yes	Yes
Second homes	No	Yes	Yes	Yes
Rental dwellings	No	No	Yes	Yes
Empty dwellings	No	No	No	Yes
Observations	16309325	19251902	26352808	29400157

*Signif. Codes: ***: 0.001, **:0.01, *:0.05*
Robust standard errors in parentheses

Table 1: Income gap between exposed and non-exposed areas

Notes. The table displays the beta coefficients of the regression $Log(Income_i) = \alpha + \beta Risk_i + \varepsilon_i$ detailed in Section 4.1. Panel A shows the results for five different regressions corresponding to each type of dwelling. For rental dwellings, I measure the income gap both for landlords and tenants. Panel B displays the income gap for owners adding successively the different types of dwellings.

Panel A: Income gap by type of dwelling

	Owner's log living standard				Tenant's log living standard
	(1)	(2)	(3)	(4)	(5)
Dwelling located in a risky area	-0.053***	-0.020	-0.023*	-0.023	-0.039***
	(0.008)	(0.013)	(0.010)	(0.013)	(0.010)
Owner-occupied dwellings	Yes	No	No	No	No
Second homes	No	Yes	No	No	No
Rental dwellings	No	No	Yes	No	Yes
Empty dwellings	No	No	No	Yes	No
Municipality fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	3658039	600856	2281268	820143	2281268

*Signif. Codes: ***: 0.001, **:0.01, *:0.05*
Standard errors clustered at the municipality level in parentheses

Panel B: Aggregate owners' income gap

	Owner's log living standard			
	(1)	(2)	(3)	(4)
Dwelling located in a risky area	-0.053***	-0.026**	-0.021**	-0.020**
	(0.008)	(0.009)	(0.007)	(0.007)
Owner-occupied dwellings	Yes	Yes	Yes	Yes
Second homes	No	Yes	Yes	Yes
Rental dwellings	No	No	Yes	Yes
Empty dwellings	No	No	No	Yes
Municipality fixed effects	Yes	Yes	Yes	Yes
Observations	3658039	4258895	6540163	7360306

*Signif. Codes: ***: 0.001, **:0.01, *:0.05*
Standard errors clustered at the municipality level in parentheses

Table 2: Within-municipality income gap between exposed and non-exposed areas

Notes. The table displays the beta coefficients of the regression $Log(Income_i) = \alpha + \beta Risk_i + \gamma_m + \varepsilon_i$ detailed in Section 4.1. γ_m correspond to municipality fixed effects. Only municipalities with at least one dwelling exposed are included in the regression. Panel A shows the results for five different regressions corresponding to each type of dwelling. For rental dwellings, I measure the income gap both for landlords and tenants. Panel B displays the income gap for owners adding successively the different types of dwellings.

Income Percentiles	Owner-occupied dwellings	Rented Homes	Secondary Homes	Empty Dwellings	**Total**
Bottom 50%	18%	5%	3%	2%	28%
Middle 40%	26%	9%	8%	4%	47%
Top 10%	9%	8%	6%	2%	25%
Total	53%	22%	17%	8%	**100%**

Table 3: Cost of the buyout policy by income and wealth groups

Notes. The percentages in the table correspond to the cost of buying back all exposed dwellings in a given category as a share of the cost of buying back all exposed dwellings. E.g., if the government opts to purchase all at-risk owner-occupied homes belonging to the bottom 50%, the cost would only amount to 18% of the total buyout expense.